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Ethical Data Analysis using SAS Viya

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Abstract

This hands-on workshop explores predictive modeling and ethical decision-making using a guided scenario in SAS® Model Studio. You'll learn how bias and fairness issues can arise in real-world analytics, and how SAS® Model Studio's built-in fairness and bias assessment tools help identify and mitigate them. Emphasis will be placed on refining models to balance predictive performance with equitable outcomes.

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Part 1: Business as Usual

Predicting Graduate School Admissions

Welcome to iLink University! As a **Predictive Modeler** in the Admissions Department, you'll help decide who earns a spot in the Fall 2026 M.S. in Analytics Program. You'll start by exploring the tools that make this possible: the **Visual Data Mining and Machine Learning** capabilities in SAS Model Studio.

In this first part, you'll follow the same "business as usual" process the Admissions Department has used for years — using data from past applicants to predict who should be accepted next. You'll model the binary variable **Admitted,** where 1 = "Yes" (the student was offered admission) and 0 = "No" (the student was not offered admission). This variable is the **target** for your predictive models. Its Yes/No binary nature makes it ideal for a range of machine learning algorithms, such as logistic regression and decision trees.

Later, in Part 2, we'll step back and ask a deeper question: *Just because we can build a predictive model, should we?* You'll then use SAS Model Studio's Fairness and Bias assessment tools to evaluate your models and explore how to make admissions decisions that are both accurate and equitable.

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Data Dictionary

For our modeling, we have 1,000 admission decisions from the past 5 years at iLink University. Let's understand the data a bit better, by examining the data dictionary below. Admitted is the outcome variable that we really care about. As noted, it is a binary variable that can be considered a yes/no variable. The other variables can be used to help us select the top candidates for admissions in the incoming class, based upon historical precedent.

The table contains 12 columns and 1,000 rows.

Column Name	Description	Definition
Admitted	Admitted (1=Yes)	When Admitted = 1, the student is offered admission into the iLink University M.S. in Analytics Program
Analytics_Work_Experience	Analytics Work Experience	Years of analytical experience
Country_Region	Country Region	Region of the world applying from
Cultural_Identity	Cultural Identity	Cultural identity
Gender	Gender Identity or Gender at Birth	Gender identity or gender at birth
ID	Application ID	Application ID
Legacy_Admission	Legacy Admission (1=Yes)	Legacy admission = 1 if either: (a) the student's parents attended the university or (b) they previously completed another degree at iLink University.
Mission_Statement	Mission Statement	Optional mission statement (maximum of 100 words)
Standardized_Test_Score	Standardized Test Score	Standardized test score (Z-score)
Strength_of_Recommendation	Strength of Recommendations	Overall strength of recommendations (0 to 5, higher is better)
Undergrad_Degree	Undergraduate Degree Category	Area of study of undergraduate degree
Years_Work_Experience	Years Work Experience	Total years of work experience, all fields

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With a better understanding of the data, the rest of the project flows with steps the data analyst in you (generally) knows by heart:

- Create a SAS Model Studio project.
- Explore the data and get to know it A LOT better.
- Run a bunch of predictive models using the Visual Data Mining and Machine Learning tools in SAS Viya.
- Find the best model predicting the historical sample. Crown it the champion.
- Finally, you'll apply that champion model to a new set of candidates and choose 40 students that will be admitted as part of the incoming class of 2026. They will all accept – because our program is awesome.

Create a SAS Model Studio Project

- From the **Applications menu,** select **Build Models** to navigate to SAS Model Studio.
- Select **New Project** from the top right of the screen.
- The **New Project** window will step you through setting up a new project. Fill in the following:

  Name: *Admissions Analysis*

  Type: keep the default selection of *Data Mining and Machine Learning*

  Template: click **Browse** and select *Intermediate template for class target*, and **OK**

  Data: click **Browse** and select **SIMULATED_ADMISSIONS** and **OK**

  Folder: click **Browse** and select **My Folder** and **OK**.
- Once you've entered the setting, click **Save**. Your SAS Model Studio project is live and ready to explore!

Explore the Data

The **Data** tab is where every great analysis begins. It's the perfect place to get to know your data a bit better.

- Examine all the attributes for the variables on the Data tab.

  For example, check out **Admitted**, our target variable. From the summary statistics, we can see that *Admitted* has two values (0/1) with an average of 0.2 or 20%. That means 20% of the applicants are admitted to the iLink University. That's helpful information!
- The warning at the top of the Data tab states that a variable must be assigned as Target to proceed.
- Select **Admitted**, ensure that the **Properties** pane is open on the right, and change the **Role** to *Target*.

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The warning at the top of the screen should disappear.

- Now, spend a bit more time examining the other variables, just so you get a little more comfortable. Nothing else needs adjustment right now.
- Navigate to the **Pipeline** tab to reveal your **SAS Model Studio** pipeline. Now, we could simply click **Run pipeline** and go straight to the results for the three models included in our Intermediate Template (Stepwise Logistic Regression, Forward Logistic Regression, and Decision Tree). But before that, let's explore the data a bit more deeply.
- Expand the **Nodes** icon on the left pane. Take a moment to browse the available options. Then, under **Miscellaneous**, find the **Data Exploration** node and drag and drop it on top of the **Data** node in the pipeline.
- Right-click on the **Data Exploration** node and select **Run**.
- When the green checkmark appears, right-click on the node again and select **Results**.

  A new window opens with a wealth of summary statistics to help you get better acquainted with your data. You'll find metrics like missing values, correlations, interval moments, and more. Spend a few minutes reviewing the output.
- When you're done exploring, click **Close** to return to your pipeline.

Run Predictive Models

- Navigate to the **Pipeline 1** tab, click **Run pipeline** to execute the models with their default settings.
- Once the pipeline has finished running, a green checkmark will appear at the bottom of each individual node.
- To explore the results of an individual model, right-click on the node and select **Results**.
  - The **Node** tab includes traditional statistical measures such as parameter estimates, selection summaries (if applicable), fit statistics, and score code.
  - The **Assessment** tab displays an interactive dashboard packed with model-fit statistics and tools.
- Spend a little time exploring, then click **Close** when you're done reviewing.

Crown a Champion Model

- Right-click on the **Model Comparison** node and select **Results**.

  The default selection criterion for classification targets is the **KS (Youden)** statistic, and the model with the highest KS (Youden) will be crowned the champion model.

  You can also explore the **Assessment** tab to compare additional performance metrics and validation charts.

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- When you're satisfied, click **Close** to return to the pipeline.
- Navigate back to the **Pipeline 1** tab, expand the **Supervised Learning** node to explore all the modeling options available to you.

  Alternatively, right-click on the **Data** node inside your pipeline, and select **Add child node** > **Supervised Learning** > **Gradient Boosting**.
- Click **Run pipeline** to accept the modeling defaults.
- The **Pipelines 1** tab should contain multiple models ready for comparison, along with a champion identified under **Model Comparison.** To improve our results, we can:
  - Add more models to the existing Pipeline 1 template
  - Add **Text Mining** as a child node to the **Data** node to see if *Mission Statement* has any predictive power (hint, hint: it should).
  - Try using *Advanced Template for class target* and further refine those models

Select the Incoming Class of 2026

The big moment is here! You've built your models, compared their performance, and crowned a champion. Now it's time to **put that model to work** and select our next class of students.

In predictive modeling terms, this step is called **scoring** — using a trained model to generate predictions for new data. In our case, we'll use data from the current pool of applicants to identify those most similar to past successful students, and then select the **top 40** candidates for admission.

- Find your champion model in its respective pipeline.
- Right-click on the champion model node and select **Add child node** > **Miscellaneous** > **Score Data**.
- Select the Score Data node on your pipeline and expand the **Node options** on the right.
- To start, we need to pull the data on the 200 students who've applied to be part of our 2026 cohort. Make the following changes to the settings:

Score Data	**Table name**: click **Browse** and select **SIMULATED_ADMISSIONG_SCORING** for the Available data sets
Output Data	**Output library:** click **Browse** and select your CASUSER folder **Table name: Class_of_2026** ☑ **Replace existing table** ☑ **Promote table**

- Right-click on the **Score Data** node and select **Run**.

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- When the node finishes running, right-click on the node and select **Results**. Navigate to the **Output Data** tab and click **View output data**.
- Almost there! A **Sample Data** window pops up. Since we had 200 applicants, let's check them all out. Skip the sampling and click **View output data.**
- Once the data are loaded, we have a sortable data set for viewing. You can expand the view and then scroll to the right. Finally, sort – descending – on the **Predicted: Admitted=1** column.
- The **top 40 candidates**, sorted by *Predicted: Admitted=1* will make up our incoming class of 2026.

  Congratulations — you just used predictive modeling to select a new cohort of students! Let's get that list over to HR and get those letters ready!
- Click **Close** to wrap up Part 1: *Business as Usual*… or keep the momentum going with the **Student Admissions** dashboard in SAS Visual Analytics.

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Part 2: Business (not) as Usual

From your graduate studies, you already know that using historical data to predict new cases — especially for decisions like admissions — can unintentionally **perpetuate the status quo**. That might be fine in an equitable world, but it becomes a real problem when it allows **bias** to persist.

Recently, the iLink University Admissions Department recognized that our "business as usual" approach may include some bias. Until now, we've mostly relied on instinct or "gut feelings" to spot it. But we can do better.

Fortunately, **SAS Model Studio** includes powerful **Fairness and Bias assessment** tools that allow us to evaluate potential bias **quantitatively** — not just intuitively. These tools help us identify which model inputs (or variables) may be introducing unintended bias and guide us toward more equitable modeling decisions.

Starting from the Beginning

To use the fairness and bias tools in SAS Viya, let's go back to our **Admissions Analysis** project in SAS Model Studio and take it from the top:

- Navigate back to the **Data** tab and select the following variables:
  - County_Region
  - Cultural_Identity
  - Gender
  - Legacy_Admission
- On the **Properties** pane, check the box for **Assess this variable for bias**.
- Your **Data** tab should now show four check marks under *Assess for Bias* for the variables we selected. Nice work — your metadata setup is complete!

Revisit our Modeling

With bias assessment activated, it's time to rerun your models and explore the new results.

- Go to the **Pipelines** tab. You'll notice the modeling nodes have reset — that's expected. The system updates automatically to include fairness and bias analysis. Click **Run pipeline**.
- Once the pipeline finishes, identify the champion model, right-click the node and select **Results**.
- A new tab is now available for **Fairness and Bias** assessment.

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- Expand the **Performance Bias Parity** object. The note on the right reads:

  Performance bias is a representation of how accurate the model is for one group over another. The bars in this plot represent the maximum pairwise difference in performance metrics between levels of each Assess for Bias variable for the TEST partition. The performance metrics included here are Multiclass Log Loss (MCLL), True Positive Rate (TPR), and Best Kolmogorov-Smirnov along ROC (maxKS). You can view the values of the performance metrics for each level of each Assess for Bias variable in the Performance Bias plot.

  Simply put: There are several incredibly important factors driving admissions selection. We could be unconsciously weighing those factors a little too strongly in our admissions modeling.

- Expand the **Performance Bias** object, which allows us to do a deeper dive into performance bias at the variable level.

  Performance bias is a representation of how accurate the model is for one group over another. The bars in this plot represent the value of performance metrics for each level of the chosen variable for the TEST partition. The performance metrics included here are Multiclass Log Loss (MCLL), True Positive Rate (TPR), and Best Kolmogorov-Smirnov along ROC (maxKS). Large differences in bar size for any specific metric indicates that the model performance (accuracy) is not consistent across all levels of the chosen variable, and you should be aware of this before using your model.

- Next, let's expand the **Prediction Bias Parity** object.

  Prediction bias represents how much greater the model's probability to predict the event is for one group over another on average. The bars in this plot represent the maximum pairwise difference in the target event's average predicted probability between level of each Assess for Bias variable for the TEST partition. You can view the target event's average predicted probability for each level of each Assess for Bias variable in the Prediction Bias plot.

  Yikes! While we can't tell the direction, there are several factors that seem to lead to big swings in acceptance rates.

- Expand the **Prediction Bias** object.

  Prediction bias represents how much greater the model's probability to predict the event is for one group over another on average. The bars in this plot represent the target event's average predicted probability for each level of the chosen variable for the TEST partition. Large differences in bar size indicate that the model predicts the event at considerable different rates for different levels of the chosen variable, and you should be aware of this before using your model.

  There are several factors that we should drop from the analysis because they potentially perpetuate historical trends in acceptance, rather than the aptitude of the students, and ultimately we want the best students.

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- To examine other variables in the **Fairness and Bias** output, use the dropdown boxes in each object.

  Spend a bit of time exploring **the Fairness and Bias** results and consider:

  - Which variables are just perpetuating historical trends in admissions – rather than helping us accept the best students?
  - And, as such, which variables should we drop to create a better admissions process?

- When you're done exploring, click **Close** to return to your pipeline.

Rerunning the Entire Analysis

Modeling is an **iterative process** — we rarely get everything right the first time. Each model builds upon what we learned from the last. **SAS Viya** automates much of the re-modeling process for you. Let's look at a few quick ways to update your analysis and keep improving your model.

It's easy to adjust your metadata

Updating your metadata — and re-running your models — takes just a few clicks.

- Return to the **Data** tab.
- Identify and select any variables that appear to be introducing bias.
- From the **Properties** pane, change the **Role** from *Input* to *Rejected.*
- Once variable metadata is changed, the pipeline resets and needs to be run again. Select **Run pipeline** and explore!

It's easy to try new models

We really want to capture as many factors as possible and that means finding a way to weigh several factors into a decision process. You did this in *Part 1*, but if you've dropped variables then the data have changed and you may need to work a little harder to find information for the algorithms to use to predict new cases based upon a smaller selection of input variables.

- Return to your **Pipeline** and consider adding **Data Mining Preprocessing** nodes!

  Why would we want to do this? Well, we have a limited amount of information on each candidate – but want to select the best overall candidates. Towards that end, we can create new variables from existing variables – to account for things like interactions between factors (like standardized test scores and work experience) or we could actually use *Mission Statement*, which is currently being ignored.

- **Ctrl-Click** the paths that you want to be connected to the **Text mining** node.

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- With those paths identified, right-click, select **Insert** > **Data Mining Preprocessing** > **Text Mining.**
- Click **Run pipeline.** This pipeline now (1) has some text analytics variables and (2) excludes questionable demographic variables that may have unwittingly introduced bias into my modeling. Know that each iteration brings you closer to a model that is both **empirically strong and ethically sound**.
- Once the pipeline is ready, view **Results** of the **Model Comparison** node to identify your champion model.
- If needed, move the **Score Data** node to the new champion model and keep the settings the same (I recommend that you copy-and-paste… and then delete the old node).
- **Run pipeline** to score using the new champion model! Check marks the spot!

Let's take a moment to celebrate all you've accomplished! Here's a quick recap of your journey:

- You grabbed historical data on admissions and built some models
- The algorithm gave us a list of applicants… and we could have just run with it.
- But instead, we used Fairness and Bias assessment tools in SAS Model Studio to create a more equitable admissions process – one that better seeks to accept the absolute best candidates rather than the candidates that mirrored the successful applicants of our recent past.

This exercise hopefully highlights an important point:

> *Algorithms just do what they are told (i.e., maximizing or minimizing some fit statistic). It's up to us – as humans in the loop – to determine whether the model is being used properly!*

That's why responsible modeling matters! We hope you feel confident not just running models — but thinking critically about **what those models mean.** That's the true mark of an ethical data scientist!

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